

- 5 A cell P, a fixed resistor R and a uniform resistance wire AB are connected in a circuit as shown in Fig. 5.1.

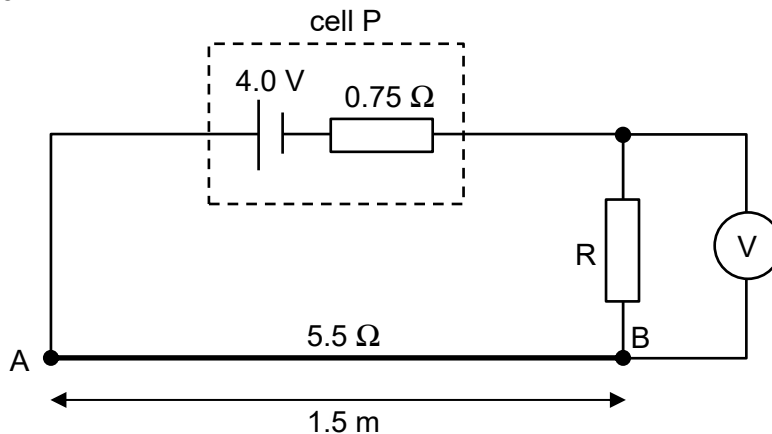


Fig. 5.1

Cell P has e.m.f. 4.0 V and internal resistance 0.75 Ω . Wire AB has length 1.5 m and resistance 5.5 Ω . The voltmeter reads 1.3 V.

- (a) Show that the potential difference across AB is 2.4 V.

- (b) A cell Q and a sensitive ammeter are connected to the circuit in Fig. 5.1, as shown in Fig. 5.2.

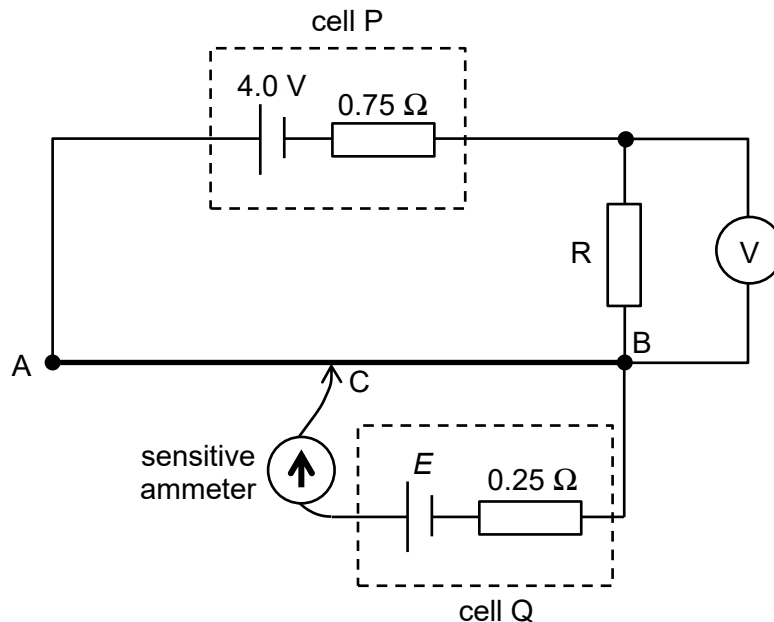


Fig. 5.2

Cell Q has e.m.f. E and internal resistance $0.25\ \Omega$. The ammeter reads zero when the length of AC is $0.56\ \text{m}$.

- (i) Determine E .

$$E = \dots\dots\dots \text{ V } [2]$$

- (ii) There is a reading on the ammeter when the connection C is shifted closer to A.

State and explain the direction of the current across cell Q.

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 [2]